RAG ASSIGNMENT EXPERIMENT REPORT

Baseline Retrieval vs. Cross-Encoder Re-Ranking

1 Experiment setup

The experiment tests whether re-ranking the same retrieved candidates improves the evidence supplied to a language model. The source is *NIST SP 800-63B: Digital Identity Guidelines—Authentication and Lifecycle Management*, June 2017 with updates through March 2020. The downloaded PDF includes its withdrawal cover and has 80 physical pages. It is used only as a fixed historical corpus; it is not current security guidance.

The first 11 PDF pages are front matter. The remaining 69 pages were cleaned of repeated headers and split into **215 page-local chunks**, each at most 220 embedding tokens, with up to 40 tokens of overlap. Stable chunk IDs retain both physical and printed page numbers. For body pages, printed page number equals physical PDF page minus 11. A SHA-256 check freezes the source PDF; the same corpus is used in both pipelines.

Role	Model / implementation
Dense embeddings	`sentence-transformers/all-MiniLM-L6-v2`
Candidate re-ranking	`cross-encoder/ms-marco-MiniLM-L-6-v2`
Answer generation	`Qwen/Qwen2.5-1.5B-Instruct`, locally hosted
Vector store	FAISS `IndexFlatIP`; 384-dimensional normalized vectors

2 Architecture comparison

A **Bi-Encoder** embeds each chunk independently of the question. Document vectors are computed once and stored. At query time, the question is embedded and compared with these vectors. With unit-normalized vectors, the inner product equals cosine similarity. This is fast, but compressing a passage into one vector can blur a prohibition, exception, or numeric condition.

A **Cross-Encoder** processes the question and one candidate passage together. Joint self-attention lets question tokens interact directly with passage tokens before a relevance score is produced. Each of the ten pairs is scored separately, then the candidates are sorted by score. These scores are not on the same scale as cosine similarities.

The baseline retrieves ten candidates and sends its original Top 3 to the LLM. The advanced pipeline reuses those exact ten candidates and sends the re-ranked Top 3. Both use the same answer model, prompt template, context formatting, greedy decoding, and 384-token output limit. Gold labels and expected answers are not included in the prompts.

3 Evaluation design

The 32 exploratory questions cover keyword distractors, exceptions, negation, and document-specific distinctions. An exact critical phrase and page identify the evidence to inspect. The best rank among chunks containing that phrase is recorded. Six questions promoted critical evidence from ranks 4–10 into the Top 3, one demoted it out of the Top 3, and seven lacked it in the initial Top 10. Eight cases were selected for answer generation, including six promotions and two controls. These are selected stress tests, not an unbiased accuracy benchmark; phrase labels do not exhaustively identify every relevant passage.

4 Qualitative results and rank shifts

Case	Critical passage / issue	Baseline	Re-ranked
q28	`p025_c01`: first-pet password prompts	5	1
q02	`p025_c03`: periodic change / compromise	7	3
q11	`p029_c03`: out-of-band deadline	6	1
q06	`p024_c03`: space-collapsing condition	6	1
q13	`p038_c02`: biometric failure limits	5	2
q29	`p024_c03`: spaces vs. truncation	4	1
q25	`p037_c03`: biometrics are not secrets	1	5
q09	`p028_c04`: codes on locked phones	3	1

Case 1: document-specific deadline (q11, PDF page 29)

The question asks for the deadline to finish an out-of-band authentication transaction. Baseline passages describe the transaction and unrelated session reauthentication, but omit the ten-minute deadline. Answer 1 says there is no single universal deadline in the supplied text and discusses periodic checks. Re-ranking moves `p029_c03` from rank 6 to rank 1. Answer 2 states the correct ten-minute limit and cites that chunk. This is a clear repair of an essential omission and a distinction between transaction expiry and session reauthentication.

Case 2: general advice and a mandatory exception (q02, PDF page 25)

The question asks whether passwords must change monthly when there is no leak. The baseline finds general password-usability and attack discussion. Answer 1 goes further and incorrectly claims that changes after compromise are not mandated. Re-ranking introduces the rule against arbitrary periodic changes and the exception requiring a change when compromise is evidenced. The normative chunk moves from rank 7 to rank 3; another relevant usability passage becomes rank 1. Answer 2 correctly rejects unjustified monthly rotation and recognizes the compromise exception, but could state the exception's mandatory *SHALL* more precisely.

Case 3: keyword distraction and a specific prohibition (q28, PDF page 25)

The question asks whether a password prompt may request the user's first pet's name. Baseline retrieval favors appendix passages about password predictability and composition. Answer 1 concludes "No," but infers the restriction from that general discussion. The direct rule, expressed in terms of *memorized secrets*, is buried at rank 5. Re-ranking moves it to rank 1, and Answer 2 uses the explicit prohibition. The yes/no conclusion does not change; the improvement is the evidential basis. Answer 2 still omits the requested chunk-ID citation.

Remaining failures and controls

Retrieval improvements do not guarantee answer correctness. In q06, Answer 2 confuses a length the verifier should permit (64 characters) with the minimum remaining after space collapsing (8). In q13, it ignores the retrieved biometric exception: 5 consecutive failures, or 10 with qualifying presentation attack detection. In q29, it confuses Unicode normalization with the specific repeated-space allowance and still omits the eight-character condition.

Re-ranking also regresses: q25 loses the direct statement that biometrics are not secrets when its chunk drops from rank 1 to 5. The new answer keeps the correct yes/no conclusion but adds unsupported reasoning. In the negation control q09, the correct locked-screen passage was already at rank 3; moving it to rank 1 improves the justification, but is not a recovery from outside the baseline Top 3. These cases distinguish retrieval errors, grounding errors, and generation errors.

5 Latency profiling

Retrieval ran on an Apple Silicon Mac with CPU inference and four threads; answer generation used Apple MPS with float16 weights. After a warm-up, each selected question was measured five times. The table reports per-query medians in milliseconds.

Case	Dense retrieval	Additional re-ranking	Total with re-ranking
q28	5.43	85.83	91.35
q02	5.35	86.19	91.53
q11	5.53	89.30	94.83
q06	5.35	86.89	92.54
q13	5.63	84.23	89.75
q29	5.75	92.04	98.17
q25	5.75	96.38	101.68
q09	5.63	88.03	93.22

Dense retrieval includes question embedding and FAISS search. Additional re-ranking includes input checks, scoring all ten pairs, and sorting. Each total is measured from the start of dense retrieval to the end of re-ranking; it is not a cached scoring-only time. The three columns have independently calculated medians, so a row's displayed medians need not add exactly.

Across all 40 measured samples, pooled medians were **5.55 ms** for dense retrieval, **86.77 ms** for additional re-ranking, and **92.50 ms** for total retrieval with re-ranking. Thus, joint scoring dominates retrieval latency in this small corpus. Its extra cost buys improved evidence in some stress cases, not a universal quality gain.

Loading, PDF processing, and indexing are excluded from query latency. Recorded setup took approximately 0.12 s for retrieval-model loading, 1.63 s for embedding/indexing, and 2.95 s for answer-model loading, excluding downloads. LLM calls were timed once per answer with accelerator synchronization. Saved end-to-end totals are estimates combining median retrieval and a single generation call.

6 Why not apply a Cross-Encoder to every chunk?

Cross-Encoder relevance depends jointly on the new question and passage. Exhaustive scoring of N chunks therefore requires N transformer pair evaluations per query; scoring the Top 10 requires ten. For this 215-chunk corpus, exhaustive scoring means 21.5 times as many pair evaluations. This is a count comparison, not a measured runtime multiplier.

Dense search reuses stored document vectors. The exact flat index still scans them, but cheap vector comparisons replace joint transformer passes. Large collections can use approximate vector indexes. The trade-off is candidate recall: re-ranking cannot recover evidence absent from the initial Top 10, as seven exploratory questions demonstrate.

7 Scope and reproducibility

Checks of source phrases, normalization, input lengths, candidate reuse, prompts, answer completion, and timing arithmetic passed. Raw answers retain their mistakes. The small LLM, selected stress cases, incomplete relevance labels, and machine-specific timings limit broader conclusions.

Evidence: `rag_assignment.ipynb`; `outputs/candidate_ranks.csv`; `outputs/answers.json`; `outputs/latency_samples.csv`; `outputs/environment.json`; `outputs/qualitative_review.json`.

Source: NIST SP 800-63B, historical edition. The included PDF preserves its withdrawal notice.

Repository name: `shadesin/rag_assignment`.